

Valentina Astore

PHD STUDENT AT INRIA SACLAY & ÉCOLE POLYTECHNIQUE

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Age: 26 · Nationality: Italian

Education

PhD in Computer Science

Palaiseau, France

INRIA PARIS-SACLAY & ÉCOLE POLYTECHNIQUE

Oct. 2024 - Present

- Research areas: Algebraic Coding Theory, Finite Fields, Finite Geometry, and applications to Cryptography.
- Supervisors: Alain Couvreur and Martino Borello.

MSc in Mathematics (Cryptography & Coding Theory)

Trento, Italy

UNIVERSITY OF TRENTO

Sept. 2021 - March 2024

- Highly selective master program (10 students per year, with admission test).
- Graduation grade: 110 with honour/110.

Master's Project

Saint-Denis, France

UNIVERSITY OF PARIS 8 - LAGA

Oct. 2023 - Dec. 2023

- Topics: Coding Theory, Rank-Metric Codes, Distinguishers, Castelnovo-Mumford Regularity.
- Supervisors: Martino Borello (University of Paris 8) and Marco Calderini (University of Trento).

BSc in Mathematics

Trento, Italy

UNIVERSITY OF TRENTO

Sept. 2018 - Sept. 2021

- Bachelor's project: "Paillier Cryptosystem", Supervisor: Nadir Murru (University of Trento).
- Graduation grade: 108/110.

Publications

PREPRINTS

V. Astore, M. Borello, A. Couvreur, F. Salizzoni. **A q -analogue of the rational normal curve and linearized Reed-Solomon codes.** *arXiv:2606.13246*

JOURNAL PAPERS

V. Astore, M. Borello, M. Calderini, F. Salizzoni. **A geometric invariant of linear rank-metric codes.** *SIAM Journal on Applied Algebra and Geometry*. 2025

Complementary Activities

TALKS

Contributed talk. 25-29 May 2026, Naples (Italy).

Event: Combinatorics 2026.

Topics: Pure and applied combinatorics.

Contributed talk. 23-26 February 2026, Bordeaux (France).

Event: OpeRa 2026 - Open Problems on Rank-Metric Codes.

Topics: Latest advancements in rank-metric and sum-rank metric Coding Theory.

Contributed talk. 15 September - 19 September 2025, Brussels (Belgium).

Event: Finite Geometry and Friends - A Brussels summer school on Finite Geometry.

Topics: Finite geometry and applications.

Invited talk. 07-11 July 2025, Madison (Wisconsin, US).
Event: SIAM Conference on Applied Algebraic Geometry (AG25).
Topics: Algebraic geometry in industrial and applied mathematics.

Contributed talk. 31 March - 04 April 2025, Pornichet (France).
Event: Journées Codage et Cryptographie (C2).
Topics: Cryptography and Coding Theory.

Contributed talk. 14-16 February 2024, Caserta (Italy).
Event: OpeRa 2024 - Open Problems on Rank-Metric Codes.
Topics: Latest advancements in rank-metric Coding Theory.

WORKSHOPS

COSMO 2025. 03-05 February 2025, Paris (France).
Topics: Combinatorics of spaces under multiplication and applications.
Member of the organizing committee.

PQCifris 2022 School on Post-Quantum Cryptography. October 2022, Trento (Italy).
Topics: Introduction to post-quantum cryptography, multivariate-based, code-based, isogeny-based, and lattice-based methods.

De Cifris Trends in Modern Cryptography. March 2022 - May 2022, Trento (Italy).
Topics: Post-quantum cryptography, cloud encryption and secure multiparty computation.

Academic Teaching

TEACHING ASSISTANT

Introduction to Computer Programming. Fall-Winter 2025.
CSC_1F001_EP course at École polytechnique

Cryptography. Fall 2025.
CSC_5CY08_TA course at ENSTA Paris

Foundations of Computer Science: Logic, Models, Computability and Computational Complexity. Fall 2024.
CSC_41012_EP - INF 412 course at École polytechnique

Professional Experience

Substitute Teacher

SCUOLA PRIMARIA MARCONI

- On-call work
- Sector: Education (primary school).

*Bassano del Grappa (VI), Italy
May 2024*

Research and Development Programmer

BV TECH S.P.A.

- Internship experience, held remotely.
- Sector: IT Services and IT Consulting.
- Tasks: Analyze and improve the security strength of a communication service for messages and VoIP calls.

*Milano, Italy
March 2023 - July 2023*

Administrative assistant

ESSENTIAL 6 LTD

- Internship experience.
- Sector: First Aid & Safety Training .
- Tasks: Management of electronic and paper mail, production and revisions of customer's certificates, organization of team building events for the staff.

*Torquay, United Kingdom
August 2017- Sept. 2017*